Chapter 15: Architecture Maturity Models

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 15 (pp. 669-700)

Overview

Chapter 15 (pp. 669-700) explains how to measure and raise the maturity of an organization's Architecture Capability. A maturity model benchmarks architecture practice against an ideal and gives a structured path to improve (p. 669). Section 15.1 compares the two models named in the "Architecture Maturity Models" Series Guide: the **Architecture Capability Maturity Model (ACMM)**, with nine elements and six levels and an IT focus, and **Capability Maturity Model Integration (CMMI)**, with five elements and five levels and a strategic focus (pp. 670-689). Section 15.2 shows how to move up a level, with worked examples for "Strategy and vision" and "Architecture method and process" (pp. 690-695). Section 15.3 links CMMI elements to the competencies and roles from Chapter 10 (pp. 695-698). Section 15.4 summarizes (pp. 699-700). Leave with this idea: Jager prefers CMMI for EA because EA is "a strategic business management tool, not a technical instrument", and raising maturity takes several roles, not one architect (pp. 670, 698). Chapter 15 is part of the GOLD reading tour (p. 669).

> **Reading note:** Both models are referenced by the Standard through a Series Guide. The Standard has no maturity model of its own (p. 698). The preference for CMMI, the critique of ACMM's IT-only security element, the improvement activities (Tables 15-7, 15-8), and the competency mappings (Tables 15-9, 15-10) are Jager's. Don't confuse ACMM's levels (0 None to 5 *Measured*) with CMMI's levels (1 Ad hoc to 5 *Optimal*).

Detailed Section Breakdown

Chapter Opener (p. 669)

- **What a maturity model does.** It describes an ideal EA implementation, usually tied to a framework, as a benchmark. It shows current maturity and a pathway to better business-IT alignment (p. 669).
- **Two methods** in the Series Guide (ACMM, CMMI). Organizations may also build their own (p. 669).

15.1 Determining Architecture Maturity (pp. 670-689)

- **When.** Typically in the Preliminary Phase, to gauge the internal Architecture Capability (p. 670; see Ch. 12.3).
- **ACMM** measures architecture process components, often relative to IT, to find weaknesses. **CMMI** emphasizes maturing the capability to achieve organizational goals, so it is more appropriate for EA (p. 670).

15.1.1 Architecture Capability Maturity Model (pp. 670-681)

- **Three parts.** The model, the elements grouped by maturity level, and a scoring mechanism (p. 670).

Nine ACMM elements (Table 15-1)	What is assessed
Architecture process	How defined, documented, and consistently followed the process is
Architecture development	Effectiveness of the methods and tools, and adherence to standards
Business linkage	Alignment of IT investments with business objectives
Senior management involvement	Leadership sponsorship and resources

Nine ACMM elements (Table 15-1)	What is assessed
Business unit (operating unit) participation	How involved the business units are
Architecture communication	How well architecture is communicated
Architecture governance	Structures overseeing architecture
IT security	Ability to meet cybersecurity requirements
IT investment and acquisition strategy	Allocation of resources to existing and new capabilities

(pp. 671-673)

ACMM level (Table 15-2)	Meaning
0 None	No architecture at all (often omitted, since there is nothing to assess)
1 Initial	Elements recognized as worth developing. No formal processes. Ad hoc and isolated
2 Under development	Initial methods and tools. Informal, not yet consistent
3 Defined	Standardized practices, aligned with strategy, dedicated team
4 Managed	KPIs monitored. Refinement with executives, focus on ROI
5 Measured	Optimized and a reference for others. Analytics drive continuous improvement

(pp. 673-675)

- **Full framework (Table 15-3).** Describes each element at each level, e.g., Architecture process moves from "ad hoc and localized" (Initial) to "part of the culture, quality metrics captured" (Managed) to "concerted efforts to optimize" (Measured). IT investment moves to "no unplanned IT investment or acquisition activity" (Measured) (pp. 676-680).

Jager's commentary (author's view): ACMM "focuses primarily on IT architecture", so it is less suited to EA. Its "IT security" element contradicts the Standard's own view that security is a cross-cutting concern (Ch. 11.2). CMMI is "a much better fit" (p. 681).

15.1.2 Capability Maturity Model Integration (pp. 681-689)

Five CMMI elements (Table 15-4)	Meaning and practice
Strategy and vision	A clear architecture strategy and vision aligned with goals. Stakeholder involvement at all levels. Regular evaluation
Architecture governance	Competent people with clear roles and responsibilities. Senior management training, onboarding, awareness. Collaboration with similar organizations
Architecture method and process	Standardized, maintained processes improved via metrics. EA is a means to achieve goals, not an end
Architecture deliverables	Suitable tools. Artifacts used, maintained, and pruned. Integrated tools with a repository
Business alignment	Clear controls, KPIs, and quality standards. Regular stakeholder consultation. Awareness of the capability

(pp. 682-685)

CMMI level (Table 15-5)	Meaning
1 Ad hoc	No frameworks or standards. Informal. Relies on individuals. Strategy not yet translated into implementation
2 Repeatable	Formalized, repeatable process and a vision. Templates being built. Senior management recognizes the need. Metrics captured
3 Defined	Framework chosen. Formally defined process. Standards widely used. Roadmap exists

CMMI level (Table 15-5)	Meaning
4 Managed	Measurement data actively improves the process. EA informs strategy
5 Optimal	Mature. Drivers, goals, and objectives aligned with EA. No project proceeds without EA input. Collaboration beyond the organization

(pp. 685-687)

- **Full framework (Table 15-6).** E.g., Strategy and vision: ad hoc activities, then a draft vision, then deployment with defined roles, then evaluation and adjustment, then proactive action plans based on measured data (pp. 688-689).

Jager's commentary (author's view): EA's ultimate goal is to support *strategy* (Ch. 9), though in practice it may only support solution delivery or projects (p. 684).

15.2 Maturing the Architecture Capability (pp. 690-695)

- **Approach.** For each CMMI element, compare the current and next level and identify the actions needed (p. 690).
- **Example 1: Strategy and vision, Repeatable to Defined.** The vision needs governance, roles, responsibilities, and an executive sponsor. Ask: is ownership assigned at executive level? Is there a process owner? Is business management responsible? Is the owner accountable for business contribution? (pp. 691-692).

Activities (Table 15-7)	Description
Obtain mandate for architecture	Senior management commits and explicitly assigns responsibility
Create responsibility matrix	RACI mapping architecture tasks to functions
Establish architecture board	Formally approves products. Escalation platform
Appoint architecture process owner	Responsible for process effectiveness and efficiency
Assign final responsibility for architecture	Senior (especially business) management effectively involved

(p. 692)

- **Example 2: Architecture method and process, Ad hoc to Repeatable.** Ask: is architecture made only when there is a client? Is there an action plan? Is the process described and known? Is it reviewed regularly? (pp. 693-694).

Activities (Table 15-8)	Description
Plan architecture	Decide on needed products with stakeholders. Make an action plan
Describe architecture processes	Creating and maintaining architecture, formalizing products, role in projects
Establish architecture product management process	Procedures to maintain and update the architecture as a whole

(p. 694)

Five steps to raise maturity	
1	Identify the current maturity level per element
2	Determine the next desired level
3	Ask targeted questions
4	Implement the identified activities
5	Review and reassess, and raise the rating if justified

(p. 695)

- Decide actions together with senior management and key stakeholders (p. 695).

15.3 Alignment with Competencies (pp. 695-698)

- **Competencies drive maturity.** Table 15-9 maps the ten competencies from Ch. 10.4 (strategic thinking, decision-making, leadership, governance and compliance, communication, adaptability, analytical thinking, systems thinking, change leadership, business-technology alignment) to the five CMMI elements. Table 15-10 adds the core roles that hold each competency (pp. 696-697).
- **Conclusions.** One architecture role is not enough to significantly raise maturity, and the Enterprise Architect has a significant stake in professionalizing the capability (p. 698).
- **Use.** Turning maturity activities into initiatives raises maturity (p. 698).

Jager's commentary (author's view): The Standard has no maturity model of its own yet. He hopes a future one will focus on the strategic nature of the discipline rather than IT (p. 698).

15.4 Summary (pp. 699-700)

- ACMM: nine elements, IT-related, finds weaknesses. CMMI: five elements, goal-oriented, more suitable for EA (pp. 699-700).
- Competencies and multiple roles, especially the Enterprise Architect, are needed to raise maturity (p. 700).

Key Terms & Definitions

Term	Definition	Page
Maturity model	Framework benchmarking architecture practice against an ideal, with a path to improve	p. 669
ACMM	Architecture Capability Maturity Model: nine elements, levels 0-5, IT-oriented	pp. 670-674
CMMI	Capability Maturity Model Integration: five elements, levels 1-5, goal-oriented	pp. 681-686
Maturity element	An aspect of the practice assessed per level (e.g., Strategy and vision)	pp. 671, 682
Scoring mechanism	The ACMM part used to determine and report the current level	pp. 670-671
Architecture process owner	Role responsible for the effectiveness and efficiency of architecture processes	p. 692

Figures & Tables Referenced

Figure / Table	What it shows	Page
Tables 15-1, 15-2	ACMM elements and maturity levels	pp. 671, 674
Table 15-3	Full ACMM framework	pp. 676-680
Tables 15-4, 15-5	CMMI elements and maturity levels	pp. 682, 685
Table 15-6	Full CMMI framework	pp. 688-689
Figure 15-1, Table 15-7	Strategy and vision excerpt and activities	pp. 691-692
Figure 15-2, Table 15-8	Architecture method and process excerpt and activities	pp. 693-694
Tables 15-9, 15-10	CMMI elements mapped to competencies and roles	pp. 696-697

Self-Check Questions

Q: Which two maturity models does the "Architecture Maturity Models" Series Guide name?

A: The Architecture Capability Maturity Model (ACMM) and Capability Maturity Model Integration (CMMI) (p. 669).

Q: In which ADM phase is a maturity model typically used?

A: The Preliminary Phase, to gauge the Architecture Capability's maturity (p. 670).

Q: Name at least six of the nine ACMM elements.

A: Architecture process; architecture development; business linkage; senior management involvement; business unit participation; architecture communication; architecture governance; IT security; IT investment and acquisition strategy (p. 671).

Q: Name the five CMMI elements.

A: Strategy and vision; architecture governance; architecture method and process; architecture deliverables; business alignment (p. 682).

Q: Give the ACMM and CMMI level names.

A: ACMM: 0 None, 1 Initial, 2 Under development, 3 Defined, 4 Managed, 5 Measured. CMMI: 1 Ad hoc, 2 Repeatable, 3 Defined, 4 Managed, 5 Optimal (pp. 674, 685).

Q: Why does Jager prefer CMMI for Enterprise Architecture?

A: ACMM focuses on IT architecture (even treating security as IT-only), while CMMI focuses on maturing the capability to achieve organizational goals, which fits EA as a strategic discipline (pp. 670, 681).

Q: What activities move "Strategy and vision" from Repeatable to Defined?

A: Obtain a mandate, create a RACI, establish an architecture board, appoint a process owner, and assign final responsibility to (business) senior management (p. 692).

Q: What are the five steps for raising a maturity level?

A: Identify the current level; determine the next level; ask targeted questions; implement the activities; review and reassess (p. 695).

Q: What does mapping competencies to CMMI elements reveal?

A: Several architecture roles are needed to raise maturity, and the Enterprise Architect has a significant stake (p. 698).

Q: What characterizes CMMI level 5 (Optimal)?

A: A mature process aligned with drivers, goals, and objectives. Continuous improvement. No projects proceed without EA input. Collaboration with similar organizations (p. 687).